



MANIPAL SCHOOL OF INFORMATION SCIENCES

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Master of Engineering – M.E in Computer Science and Engineering

Course Plan

Course Name	:	Applied Machine Learning Lab
Course Code	:	AME 5052
Academic Year	:	2026-2027
Semester	:	I
Name of the Course Coordinator	:	Dr. Arockiaraj
Name of the Program Coordinator	:	Dr. Sathyendranath Malli

	
Date:	13-07-2026
Signature of Program Coordinator	
with Date	

	
Date:	10-07-2026
Signature of Course Coordinator	
with Date	



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Program Education Objectives (PEOs)

The overall objectives of the Learning Outcomes-based Curriculum Framework (LOCF) for Master of Engineering – M.E in Computer Science and Engineering, programme are as follows

PEO No.	Education Objective
PEO 1	Enable to draw upon fundamental and advanced knowledge to apply analytical and computational approaches to solve technological problems in computer science.
PEO 2	Introduce state of art technologies in computer science and inculcate ethical practices to make industry-ready professionals.
PEO 3	Promote scientific and societal advancement through research and entrepreneurship.



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Program Outcomes (POs)

By the end of the postgraduate program in Master of Engineering – M.E in Computer Science and Engineering, graduates will be able to:

PO 1	An ability to independently carry out research /investigation and development work to solve practical problems.
PO 2	An ability to write and present a substantial technical report/document.
PO 3	An ability to demonstrate a degree of mastery over the area as per the specialization of the program. The mastery should be at a level higher than the requirements in the appropriate bachelor program.
PO 4	An ability to develop software based on theoretical principles and practical knowledge.
PO 5	An ability to demonstrate knowledge of the underlying principles and evaluation methods for analysing and decision-making.



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1. Course Plan

1.1 Primary Information

Course Name	:	Applied Machine Learning Lab [AME 5052]
L-T-P-C	:	3-0-0-3
Contact Hours	:	36 Hours
Pre-requisite	:	Python programming fundamentals
Core/ PE/OE	:	Core

1.2 Course Outcomes (COs), Program outcomes (POs) and Bloom's Taxonomy Mapping

CO	At the end of this course, the student should be able to:	No. of Contact Hours	Program Outcomes (POs)	BL
CO1	Apply different types of supervised machine learning algorithms to practical problems.	12	PO3	3
CO2	Apply different types of unsupervised machine learning algorithms to practical problems.	12	PO4	3
CO3	Analyse different types of machine learning paradigms.	12	PO1	4



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1.3 Assessment Plan

Components	Lab test	Flexible assessments	End semester exam (ESE) / Makeup exam
Duration	90 minutes	1 (15 days)	180 minutes
Weightage	0.3	0.2	0.5
Typology of questions	Apply	Apply and analyse	Apply and analyse
Pattern	Answer all the questions. Maximum marks 30.	Answer all the questions. Maximum 20 marks.	Answer all the questions. Maximum marks 50.
Schedule	As per academic calendar.	22-Oct-26	As per academic calendar.
Topics covered	Introduction to Machine Learning, Decision Trees, Linear Models-Linear regression, Logistic regression.	Applying and analysing supervised and unsupervised machine learning models.	Comprehensive examination covering the full syllabus. Students are expected to answer all questions.



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1.4 Lesson Plan

L. No.	TOPICS	Course Outcome Addressed
L0	Course delivery plan, Course assessment plan, Course outcomes, Program outcomes, CO-PO mapping, reference books.	---
L1	Basics of Python for data handling: Introduction to Python libraries for machine learning (NumPy, Pandas, Matplotlib, Scikit-learn).	C01
L2	Understanding datasets and data preprocessing techniques such as handling missing values, feature scaling, and encoding categorical variables.	C01
L3	Implementation of decision tree classifier. Apply entropy, information gain and Gini impurity. Analyse the performance in terms of accuracy, precision, recall, F1-score. Apply cross-validation techniques.	C01
L4	Analysis of overfitting and implementation of pruning techniques. Implementation of decision tree regressor.	C01
L5	Implement linear regression model, apply different metrics (MSE, RMSE, R-sqaure).	C02
L6	Apply logistic regression model-sigmoid function, decision boundary. Regularization methods (Ridge and Lasso). Analyse the performance in terms of accuracy, precision, recall, F1-score. Apply cross-validation techniques.	C02
LT	Lab Test	0
L7	Apply K-NN algorithms for the practical problems. Analyse performance - accuracy, precision, recall, F1-score, ROC curve and AUC. Cross-validation techniques.	C02
L8	Implement maximum likelihood estimation (MLE) of parameters, Naive Bayes model for classification. Performance analysis and cross-validation techniques.	C02
L9	Computing distances and similarities, implementation and analysis of K-means clustering and Hierarchical clustering, dendrogram construction and linkage methods	C03
L10	Through coding, understand how ensemble methods in machine learning work, implementation and analysis of bagging techniques such as Random Forest and boosting algorithms.	C03



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L11	Implementation of Gaussian Mixture Models (GMM) for clustering tasks.	C03
L12	Implementation and visualization using Principal Component Analysis (PCA). Visualize, compare, and contrast different models for practical problems	C03



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1.5 References

1. G. James, D. Witten, T. Hastie, R. Tibshirani, and J. Taylor, An Introduction to Statistical Learning: with Applications in Python, Springer Texts in Statistics. Cham, Switzerland: Springer, 2023. Online resource available at <https://www.statlearning.com/>
2. L. G. Serrano, Grokking Machine Learning, S. Thrun, Foreword, Manning Publications, 2021. Online resource from Manning Publications available at <https://www.manning.com/books/grokking-machine-learning>
3. J. VanderPlas, Python Data Science Handbook, Sebastopol, CA: O'Reilly Media, 2016.
4. A Course in Machine Learning, Hal Daumé III – Online resource available at <http://ciml.info/>
5. Module: Introduction to Machine Learning (<https://www.intel.com/content/www/us/en/developer/tools/oneapi/training/academic-program/educators/intro-machine-learning-training-kit.html>).
6. Learning path: Understand data science for machine learning (<https://learn.microsoft.com/en-us/training/paths/understand-machine-learning/>).

1.6 Other Resources (Online, Text, Multimedia, etc.)

1. Web Resources: Blog, Online tools and cloud resources.
2. Journal Articles.



Semester - I						Lab: Data Science Lab			
	8-9	9-10	10-11	11-12	12-1	1-2	2-3	3-4	4-5
Monday	AML Lab								
Tuesday									
Wednesday									
Thursday									
Friday									
Saturday									



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1.8 Assessment Plan

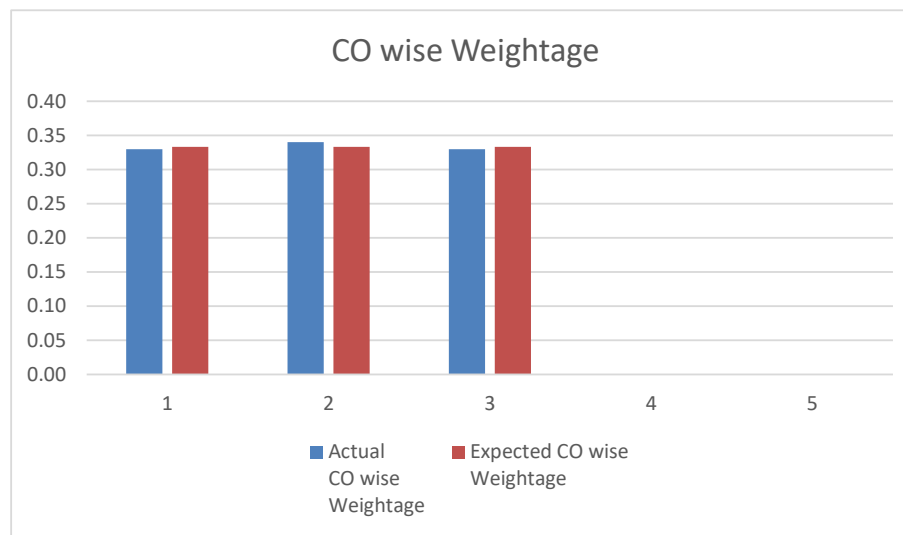
COs		Marks & Weightage				
CO No.	CO Name	Lab test marks (Max. 30)	Assignment Marks (Max. 20)	ESE Marks (Max. 50)	Actual CO wise Weightage	Expected CO wise Weightage
C01	Apply different types of supervised machine learning algorithms to practical problems.	15	5	13	0.33	0.33
C02	Apply different types of unsupervised machine learning algorithms to practical problems.	0	10	24	0.34	0.33
C03	Analyse different types of machine learning paradigms.	15	5	13	0.33	0.33
	Marks (weightage)	0.3	0.2	0.5	1.00	1.00



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Note:

- In-semester Assessment is considered as the Internal Assessment (IA) in this course for 50 marks, which includes performances in lab participation, assignment work, lab work, lab test, quizzes etc.
- ESE for this course is conducted for a maximum of 50.
- ESE marks for a maximum of 50 and IA marks for a maximum of 50 are added for a maximum of 100 marks to decide upon the grade in this course.

Weightage for CO1 = (Lab test marks for CO1 + Assignment marks for CO1 + ESE marks for CO1) /100
= (10 + 5+10)/100 = 0.25



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1.9 Assessment Details

The assessment tools to be used for the Current Academic Year (CAY) are as follows:

Sl. No.	Tools	Weightage	Frequency	Details of Measurement (Weightage/Rubrics/Duration, etc.)
1	Lab test	0.3	1	• Performance is measured using lab test attainment level. • Reference: question paper and answer scheme. • The lab test is assessed for a maximum of 30 marks.
2	Assignments	0.2	1	• Performance is measured using assignments attainment level. • Assignment is evaluated for a maximum of 20 marks.
3	ESE	0.5	1	• Performance is measured using ESE attainment level. • Reference: question paper and answer scheme. • ESE is assessed for a maximum of 50 marks.

1.10 Course Articulation Matrix

CO	PO1	PO2	PO3	PO4	PO5
CO1			Y		
CO2				Y	
CO3	Y				
Average Articulation Level	*		*	*	